Movie Recommendation Engine

Intern Details

Intern ID: CITS1931

Full Name: Boddu Neha Madhuri

Duration: 6 Weeks

Project Name: Movie Recommendation Engine

Abstract

This project presents a Movie Recommendation Engine that suggests movies based on user preferences using Content-Based Filtering. The system analyzes movie genres and computes similarity using TF-IDF vectorization and cosine similarity. It also includes fuzzy matching to handle incorrect or partial movie names. A Streamlit-based interface is developed to provide real-time recommendations.

Project Objective

- To build a movie recommendation system based on content similarity
- To analyze movie genres and compute similarity scores
- To handle user input errors using fuzzy matching
- To develop a user-friendly web interface using Streamlit

Project Scope

The Movie Recommendation Engine project focuses on developing a content-based machine learning system that suggests movies similar to a user's input. It includes data preprocessing, TF-IDF feature extraction, similarity computation, fuzzy matching, and a Streamlit web interface for real-time recommendations.

System Workflow

1. Load dataset (movies.csv)
2. Clean and preprocess movie titles and genres
3. Convert genres into TF-IDF vectors
4. Compute similarity matrix using linear kernel
5. Accept user input movie name
6. Find closest match using fuzzy matching

7. Generate top-N similar movies
8. Display results in Streamlit UI

Technologies Used

Python, Pandas, NumPy, Scikit-learn, Streamlit, FuzzyWuzzy

Methodology

Content-Based Filtering: Recommends movies based on similarity of genres.
TF-IDF Vectorization: Converts text into numerical feature vectors.
Cosine Similarity: Measures similarity between movies.
Fuzzy Matching: Handles misspelled movie names.

Dataset Description

The dataset contains movie titles and genres used to build the recommendation system.

System Implementation

The system preprocesses data, applies TF-IDF vectorization, computes similarity matrix, and uses fuzzy matching to identify closest movie titles before generating recommendations.

Result

The system successfully recommends movies based on user input with improved accuracy using fuzzy matching.

Future Scope

- Add movie posters using TMDB API
- Implement hybrid recommendation system
- Include ratings and popularity metrics
- Deploy application online

Conclusion

The project successfully demonstrates a content-based movie recommendation system with an interactive UI.